

Hivemind

Three control-plane attacks

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Last week → today

Last week's questions	Current status
Can we force leadership without knowing the group key K?	Ongoing
Crawl a live swarm every d seconds and record who has advertised.	No live averaging swarm found. Need a known group key to crawl.
Measure lookup time and the shortest advertised deadline.	Local tests using RIPE Atlas ping RTTs: Sweden, US, Brazil, India, Australia.

The delay profile: RIPE Atlas anchor RTTs

RIPE Atlas is a global measurement network whose anchor hosts continuously ping each other. Each cell is the median of three ICMP replies.

from \ to	SE	US	BR	IN	AU
Stockholm	9	99	209	147	298
New York	99	13	125	285	240
São Paulo	209	111	16	228	305
Mumbai	147	240	228	23	468
Sydney	299	239	305	503	12

233 ms

median between regions

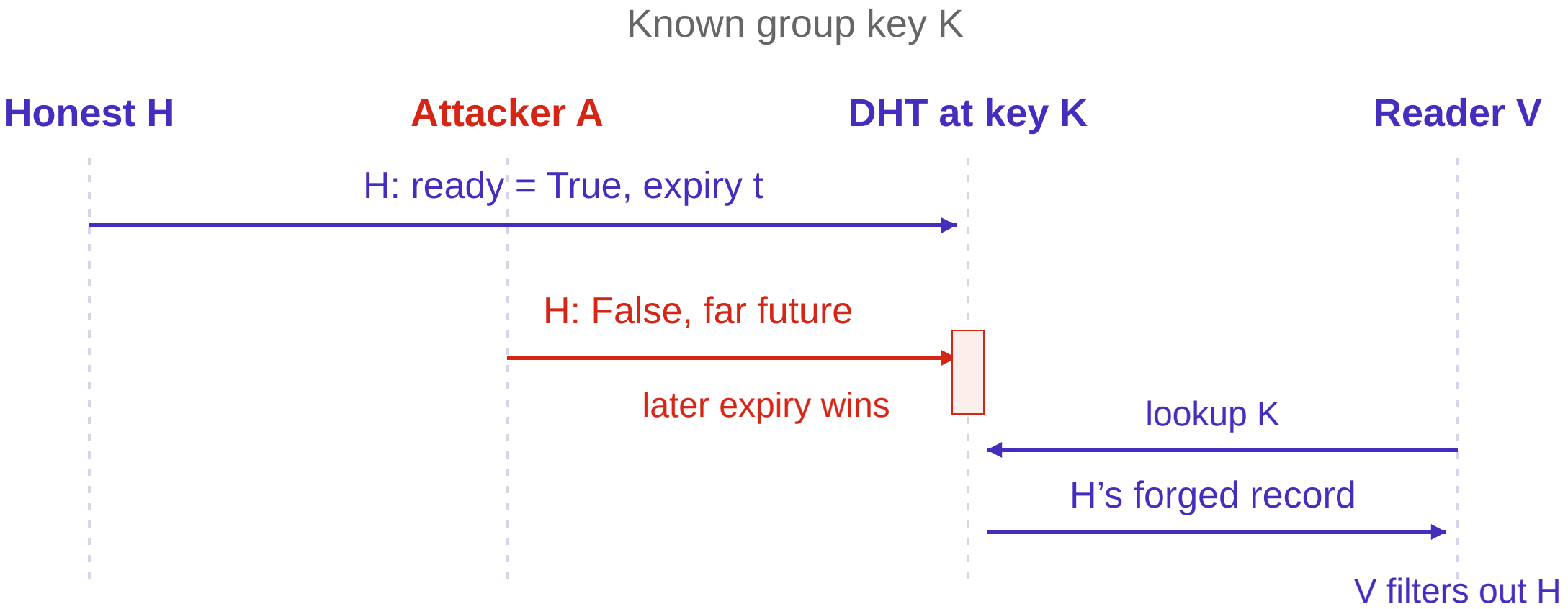
99 – 503 ms

range across the 20 links

Each region holds 5–6 of the 28 nodes in the swarm

Link RTTs, not DHT operation times · measured 30 September 2026

1. Forged withdrawal



H stays online, but disappears from group discovery.

H’s refresh expires sooner; the DHT keeps the record that expires later.

Withdrawal: two separate tests

1 Can an attacker hide honest peers?

12 or 40 honest advertisers, plus one attacker.

Check whether forged withdrawals hide peers, even after they advertise again.

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2 Does hiding a peer make the target join the attacker?

Use RIPE Atlas delays in a local swarm.

Compare who the target joins, with and without a forged withdrawal.

Withdrawal: results

Check	Observed result
Targeted peers hidden	39/39 (40 honest peers)
Refreshes that restored the peer	0/30 attempts
Target joined attacker, with withdrawal	0/10 trials
Target joined attacker, without withdrawal	0/10 trials

Group capture was tested locally with delays derived from RIPE Atlas pings.

2. Deadline inflation

Keep H visible, but move its deadline behind the attacker's.

Before

Peer	Advertised deadline
H	now + 5 s
A	now + 6 s

V would contact H

forge H's expiration

After forged write

Peer	Advertised deadline
A	now + 6 s
H	now + 1 hour

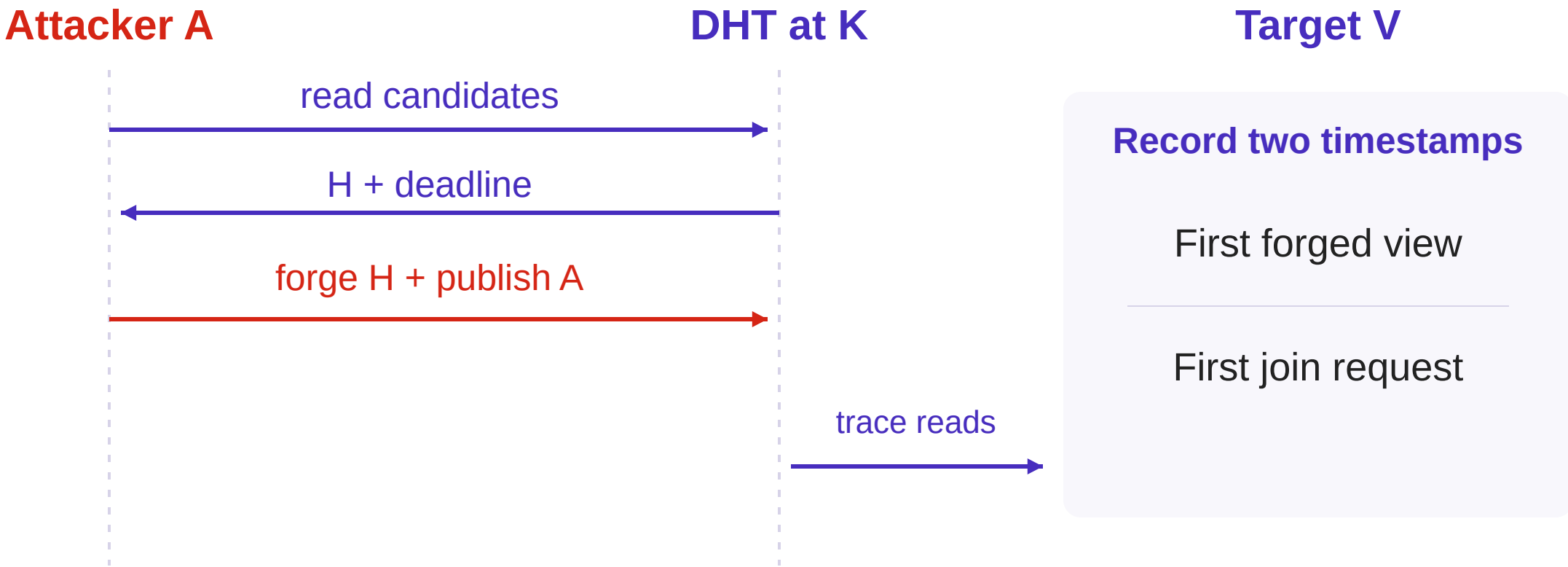
V may contact A

Only if V reads the changed record before choosing whom to join.

Illustrative times · H's ready flag stays True · no crafted DHT identity

Inflation: measure the race to V’s view

A’s own deadline stays later than H’s; only H’s record is forged.



Compare timestamps: forged view before first join request?

Accepted writes did not change V’s join

Measured event	Attack trials
DHT accepted the forged record	**10 / 10**
Forged view recorded before V’s first request	**0 / 10**
V joined the attacker	**0 / 10**

No-forgery control: 0 / 10 targets joined A.

In 7/10 trials: V picked a peer → forged view appeared later

One reactive, single-peer policy; other strategies remain untested.

3. Deadline undercutting

Read every advertised deadline, then declare one step earlier than all of them.

Advertised at K

Peer	Deadline
H1	now + 11.4 s
H2	now + 12.8 s
H3	now + 13.8 s
A	now + 11.4 s − one float step

undercut the earliest →

Leader election

earliest deadline wins

A leads; A admits whom it likes

Cost to A: one identity,
no compute, no crafted node ID

Undercutting: two experiments

Same key, node IDs and honest schedule; only the attacker's region changes.

Read ads at K → **Undercut earliest** → **Publish own ad** → **Wait for joins**

Experiment 1 · where the attacker sits

5 attacker origins × 2 honest windows

3 seeds each, so 30 runs in total

A seed sets the node IDs and the honest deadline schedule

Experiment 2 · is the failure ours?

It ignores ads with under 3.5 s left

1.0 s of that is padding we chose

Set our padding to zero and retry

from Stockholm, the cheapest origin

28 local DHT nodes, 5–6 per region

Injected delays on one machine · group key known to the attacker

The window decides, not the attacker

Attacker origin	11–15 s window	5–7 s window
Stockholm	**3 / 3**	**0 / 3**
New York	**3 / 3**	**0 / 3**
São Paulo	**3 / 3**	**0 / 3**
Mumbai	**3 / 3**	**0 / 3**
Sydney	**3 / 3**	**0 / 3**

5–7 s is Hivemind's shipped default.

4. One group, conflicting reducer indices

Precondition: A is already the group leader.



different
ordered rosters

Same group ID
Same four peers

Reducer index

	0	1	2	3
A's own view	A	F1	F2	F3
Sent to F1	F1	A	F2	F3
Sent to F2	F2	F1	A	F3
Sent to F3	F3	F1	F2	A

Each recipient claims A's reducer slot (index 0 in this example).

Equal-sized slices can be mixed; followers do not cross-check one common roster.

Roster test: change only the ordering

1 Same inputs

A	0	1	2	4
F1	0	1	2	4
F2	0	1	2	4
F3	0	1	2	4

Four real local averagers

2 Two conditions

One common roster

5 paired trials

Different roster per peer

3 Check outputs

Expected average

0	1	2	4

Did it complete?

Is the tensor correct?

Were failures recorded?

Real DHT + join streams + stock all-reduce · leadership arranged in advance

Wrong tensors, successful steps

One recorded trial

Correct average

0	1	2	4

One attacked peer’s output

1.75	0.75	1.5	3.0

Across five paired trials

Corrupted trials

Control	**0 / 5**
Attack	**5 / 5**

20/20 attacked outputs wrong

All 40 steps completed; tested failure indicators stayed clear.

Tensor corruption measured locally; downstream training damage was not tested.